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CHE-210 HW set 10

1a.) This reaction is endothermic b/c $\Delta \hat{H}^\circ_{rx} > 0$ and we would have to heat up the reactor in order to keep the temp constant. In a adiabatic reaction, the rate of heat transfer is 0. As the reaction starts to absorb energy, the temp will decrease. As this is a endothermic reaction, the energy that is required to break the molecular bonds is higher than the energy that is released when the bonds are formed.

1b.) $\Delta \hat{U}_r(T) = \Delta \hat{H}_r(T) - RT\left(\sum \nu_i - \sum \nu_i\right)$

$= -69.36 \frac{kJ}{mol} - 8.314 \frac{J}{mol \cdot K}\left(\frac{1\,kJ}{1000\,J}\right)(298\,K)(7-0)$

$\boxed{\Delta \hat{U}_r = -52.017 \frac{kJ}{mol}}$

1c) $Q = \Delta U = n\Delta \hat{U}^\circ_r$ $\quad n = \frac{\text{mass of } CaCl_2}{\text{Molar Mass of } CaCl_2} = \frac{150\,g}{64.1} = 2.34$

$2.34\,mol \cdot -52.015 \frac{kJ}{mol} = \boxed{Q = +121.72\ kJ \text{ to the system}}$

6.31) $q_{water} = m C_p \Delta T$

$q_{water} = (650)(4.184)(36.2-23.6) = 3.426 \cdot 10^4 J = 34.26\,kJ$

$q_{rxn} = -34.26\,kJ$

Heated Released per g

$\frac{q_{rxn}}{m_{sample}} = \frac{-34.26}{2.7} = -12.69 \frac{kJ}{g}$

moles burned $n = \frac{2.7}{154} = 0.01753\,mol$

Table B.3 $\Delta \hat{H}^\circ_f(CO_2) = -393.5$

$\Delta \hat{H}^\circ_f(H_2O(l)) = -285.8$

$\frac{-2622}{154} = \boxed{-17.0 \frac{kJ}{g}}$

$C_8H_6O_3 + \frac{17}{2}O_2 \rightarrow 8CO_2 + 5H_2O$

$MW_{C8H6O3} = 154 \frac{g}{mol}$

Combustion enthalpy per mol

$\Delta H_{comb} = \frac{-34.26}{0.01753} = -1954 \frac{kJ}{mol}$

$\Delta H_{comb} = [8(-393.5) + 5(-285.8)] - \Delta H_r(?)$

$-1954 = (-4576) - \Delta \hat{H}_r(?) \rightarrow \Delta H_r(?) = -2622 \frac{kJ}{mol}$

P6.34 $Q = (100\ \frac{kJ}{kg})(6\ kg) = 600\ kJ$

Heat released per mole of Fe $\quad \Delta\hat{H}^\circ_{f(Fe_2O_3)} = -839.5$

$4Fe + 3O_2 \rightarrow 2Fe_2O_3$

$\Delta H_{rxn} = 2(-830.5) - 0 = -1661\ kJ$

$\frac{1661}{4} = 415.25\ \frac{kJ}{mol}\ Fe$

Moles of Fe required $\quad n_{Fe} = \frac{600\ kJ}{415.25} = 1.445\ mol\ Fe$

Mass of Fe required $\quad MW_{Fe} = 55.85\ \frac{g}{mol}$

$m_{Fe} = 1.445(55.85) = \boxed{80.7\ g}$

Mass change of the hot pack $\quad n_{O_2} = \frac{3}{4} n_{Fe} = \frac{3}{4}(1.445) = 1.08375\ mol$

$m_{O_2} = 1.08375(32) = \boxed{34.68\ g}$

Percent mass change

$m_{final} = 80.7 + 34.68 = 115.38\ g$ $\quad \%\Delta m = \frac{34.68}{80.7} \cdot 100 = \boxed{42.9\%}$

4a) $N_2 + O_2 \rightarrow NO$ $\quad \Delta H^\circ_f[NO_{(g)}] = 40.25\ \frac{kJ}{mol}$ Table A.3

$\Delta H^\circ_r = 2(90.25) - [0+0] = \boxed{+180.5\ \frac{kJ}{mol}}$

4b) gas n-pentane + Oxygen → Carbon monoxide + Water

$C_5H_{12(g)} + \frac{11}{2}O_{2(g)} \rightarrow 5CO_{(g)} + 6H_2O_{(l)}$

$\Delta\hat{H}^\circ_f[C_5H_{12(g)}] = -146.76\ \frac{kJ}{mol}$

$\Delta\hat{H}^\circ_f[CO_{(g)}] = -110.53\ \frac{kJ}{mol}$ Table B.7

$\Delta\hat{H}^\circ_f[H_2O_{(l)}] = -285.84\ \frac{kJ}{mol}$

$\Delta H^\circ_r = [5(-110.53) + 6(-285.84)] - [(-146.76) + 0]$

$= -2267.64 + 146.76 = \boxed{-2120.93\ \frac{kJ}{mol}}$

4c) liquid n-hexane + oxygen → Carbon dioxide + water vapor

$C_6H_{14(l)} + \frac{19}{2}O_{2(g)} \rightarrow 6CO_{2(g)} + 7H_2O_{(g)}$

$\Delta\hat{H}^\circ_f[C_6H_{14(l)}] = -198.8\ \frac{kJ}{mol}$

$\Delta\hat{H}^\circ_f[O_{2(g)}] = -393.5\ \frac{kJ}{mol}$

$\Delta\hat{H}^\circ_f[H_2O_{(g)}] = -241.83\ \frac{kJ}{mol}$

$\Delta H^\circ_r = [6(-393.5) + 7(-241.83)] - [(-198.8) + 0]$

$= -4053.81 + 198.8 = -3855.01\ \frac{kJ}{mol}$

Hess's Law Form

$6C\ (graphite) + 7H_{2(g)} \rightarrow C_6H_{14(l)}$

$C\ (graphite) + O_{2(g)} \rightarrow CO_{2(g)}$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}$

$\Delta H^\circ_r = \sum \nu_p \Delta H^\circ_{f,p} - \sum \nu_r \Delta H^\circ_{f,r}$

4d) liquid Sodium sulfate + Carbon monoxide → liquid sodium sulfide + Carbon dioxide

$\Delta H_f^\circ[CO(g)] = -110.53 \frac{kJ}{mol}$ Table $\Delta H_f^\circ[Na_2SO_4(l)] = -1356.38 \frac{kJ}{mol}$ NIST

$\Delta H_f^\circ[CO_2(g)] = -393.5 \frac{kJ}{mol}$ P.3 $\Delta H_f^\circ[Na_2S(l)] = -366.10 \frac{kJ}{mol}$ NIST-JANAF

$Na_2SO_4 + 4CO \rightarrow Na_2S + 4CO_2$

$\Delta H_r^\circ = [(-366.10) + 4(-393.5)] - [(-1356.38) + 4(-110.53)]$

$= -1940.10 - (-1748.50) = \boxed{-141.6 \frac{kJ}{mol}}$

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P6.37) How much reacts?

1mL $H_2O \approx 1.0$ g water $n_{H_2O} = \frac{1.0}{18.015} = 0.055 mol$

Sodium Consumed $n_{Na,reacted} = 0.0055$

$Na : H_2O = 1:1$ $m_{Na,reacted} = 0.0055(22.99) = 1.28 g$

Hydrogen produced $n_{H_2} = \frac{0.055}{2} = 0.0278 mol$ $V_{H_2} \approx 0.0278(24.5) = 0.68 L$

0.7L of hydrogen gas

Sodium hydroxide produced $n_{NaOH} = 0.055$

$m_{NaOH} = 0.055(40.0) = 2.2 g$

What else happens? $\Delta H_{rxn}^\circ = [2\Delta H_f^\circ(NaOH,aq) + \Delta H_f^\circ(H_2)] - [2\Delta H_f^\circ(Na) + 2\Delta H_f^\circ(H_2O,l)]$

$\Delta H_f^\circ(NaOH,aq) = -469.15$ $\Delta H_{rxn}^\circ = 2(-469.15) - 2(-285.84) = -366.62 kJ$

$\Delta H_f^\circ(H_2O,l) = -285.84$ $\frac{366.6}{2} = 183.3$ $q = 0.055(183.3) = 10.2 kJ$

Why is it dangerous? To heat 1g of water from 25°C to 100°C

$q_{Na} \approx 2.26$ kJ $q = mc\Delta T = (1)(4.184)(75) \approx 314 J = 0.314 kJ$

$0.314 + 2.26 = 2.57 kJ$

Explanation

- Forms hot, corrosive sodium hydroxide
- Releases enough heat to boil water instantly
- Can melt and scatter sodium
- Can ignite hydrogen

Adding water to pure sodium is dangerous: even a 1mL drop can produce about 0.68L of H_2 and release about 10kJ of heat almost instantly

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